Appendix to Ripple: Measuring the Cost of Over-Helping in Robot Fleets

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This appendix holds the material the paper refers to but has no room for: the system prompt every model call used, the one text for each event kind, one complete live call and one veto prompt with their replies, the per-seed scores behind the paper's model table, and the per-run counts for the 40 model runs. It also gives the per-seed results for the interface, policy-writing, quiet-day, and surge-length checks. The study code is the delivery crate of the Ripple repository, and the prompt and call texts here are the files beside this document, as the crate wrote them.

I. The frozen system prompt

Every model call in the study used this system prompt. It was written once, before any scored run, and never changed. The mission file holds it.

```
You are the fleet manager for a regional blood delivery
     service. A classical dispatcher assigns each request to
     the idle vehicle at the site's own hub that has the
     stock and the range and arrives soonest; it does not
     read events. Fixed-wing aircraft drop by parachute and
     cannot land; multirotors can also land on hospital
     rooftops, which take only a landing, and have a short
     range. Each hub holds stock of each blood type. Sites
     are named by hub number and letter: 3A to 3F are hub 3's
     clinics, 3H1 and 3H2 its hospitals. You never assign
     single requests: you give orders that change what the
     dispatcher may do, where vehicles and stock are, and
     what comes first. Keep trauma and urgent deliveries on
     time, keep flights out of closed areas, and do not send
     vehicles beyond their endurance. Reply with JSON only,
     no other text.
```

II. Event texts

Each event kind has one text, written once and filled in from the event's fields. The model reads events as these sentences under *New events* and *Events still in force* (Section III). The placeholders are the hub or hubs, the site, the stock item, the area, the vehicle class, and the start and end minutes.

Kind	Text
Surge	A mass-casualty incident near {hubs}: the hospitals there are sending trauma requests beyond what those hubs can fly alone, starting at minute {from}.
Shortage	{item} has run out at {hubs} as of minute {from}. Requests for {item} at their sites can only be served from hubs that still hold it.
Closure	Airspace is closed in {area} from minute {from} until minute {to}; no flights through it.
Outage	{hubs} cannot launch from minute {from} until minute {to}; its sites must be covered by other hubs.
Weather	Weather at {hubs}: no {class} launches from minute {from} until minute {to}.
Priority	Hospital {site} takes priority over routine deliveries from minute {from} until {to}.
Rooftop closed	The rooftop at hospital {site} is closed from minute {from} until minute {to}; only drops at its ground pad are allowed.
Corridor	A medevac helicopter corridor is active in {area} from minute {from} until minute {to}; no vehicle may cross it below 150 m.

III. One complete model call

The prompt below is one call to Opus as a live manager with the exceptions view and feedback on a 16-hub day, at minute 35, and the reply that came back. The prompt is what the delivery crate wrote; nothing was removed. The order format at the end is part of every live prompt. The reply switched the hospital whose rooftop had just closed to its ground pad, then moved two fixed-wing aircraft from hub 5 to hub 3 and one from hub 7 to hub 1, two of the hubs with waiting requests and no idle vehicle, each with a return time.

```
Time: minute 35 of the day; requests stop coming at minute
     480.

New events:
- The rooftop at hospital 9H1 is closed from minute 35 until
     minute 155; only drops at its ground pad are allowed.

Events still in force:
- A medevac helicopter corridor is active in the corridor
     from (-21.5, -21.4) to (-11.6, -23.2) km, 0.5 km either
     side, below 150 m from minute 2 until minute 62; no
     vehicle may cross it below 150 m.
- Weather at hubs 6, 2, 5, 7, 10: no multirotor launches from
     minute 10 until minute 70.
- Hospital 4H2 takes priority over routine deliveries from
     minute 14 until the end of the day.
- O+ has run out at hubs 10, 6, 9, 11, 14 as of minute 21.
     Requests for O+ at their sites can only be served from
     hubs that still hold it.
- Weather at hubs 12, 8, 11, 16: no multirotor launches from
     minute 25 until minute 85.
- hub 8 cannot launch from minute 26 until minute 116; its
     sites must be covered by other hubs.
```

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- Weather at hubs 12, 8, 11, 16: no multirotor launches from
     minute 26 until minute 86.
- A mass-casualty incident near hubs 2, 1, 3, 6: the
     hospitals there are sending trauma requests beyond what
     those hubs can fly alone, starting at minute 26.
- The rooftop at hospital 15H1 is closed from minute 27 until
     minute 147; only drops at its ground pad are allowed.
- The rooftop at hospital 16H1 is closed from minute 29 until
     minute 149; only drops at its ground pad are allowed.

Your last orders:
- use the ground pad at 16H1 until minute 149: applied; drops
     at 16H1's pad, 300 m from the rooftop; any vehicle can
     fly them
- reposition 1 fixed-wing from hub 16 to hub 15 until minute
     150: refused_too_few_idle; hub 16 has no idle fixed-wing
     free to go
- reposition 1 fixed-wing from hub 12 to hub 4 until minute
     180: refused_too_few_idle; hub 12 has no idle fixed-wing
     free to go

Since your last call: 21 delivered (21 on time, 0 late).

Exceptions:
- hub 1: 7 waiting (4 trauma, 3 routine) with no idle vehicle
- hub 3: 10 waiting (8 trauma, 1 urgent, 1 routine) with no
     idle vehicle
- hub 4: 2 waiting (1 urgent, 1 routine) with no idle vehicle
- hub 6: stock low: O+ 0
- hub 8: cannot launch (outage)
- hub 9: stock low: O+ 4
- hub 10: stock low: O+ 4
- hub 11: stock low: O+ 5
- hub 14: 2 waiting (2 routine) with no idle vehicle; stock
     low: O+ 8
- hub 15: 2 waiting (1 urgent, 1 routine) with no idle vehicle
Region: 16 hubs, 25/80 vehicles idle, 50 requests waiting, 0
     late.

Orders in force:
- flights avoid the corridor from (-21.5, -21.4) to (-11.6,
     -23.2) km, 0.5 km either side, below 150 m until minute
     62
- multirotor grounded at hubs 2, 5, 6, 7, 10 until minute 70
- multirotor grounded at hubs 8, 11, 12, 16 until minute 85
- multirotor grounded at hubs 8, 11, 12, 16 until minute 86
- requests for 4H2 first until minute 480
- trauma requests first until minute 480
- urgent requests first until minute 480
- 1H1 may be served from hubs 4, 5, 7
- 1H2 may be served from hubs 4, 5, 7
- 2H1 may be served from hubs 4, 5, 7
- 2H2 may be served from hubs 4, 5, 7
- 3H1 may be served from hubs 4, 5, 7
- 3H2 may be served from hubs 4, 5, 7
- 6H1 may be served from hubs 4, 5, 7
- 6H2 may be served from hubs 4, 5, 7
- 8A may be served from hubs 4, 7, 12
- 8B may be served from hubs 4, 7, 12
- 8C may be served from hubs 4, 7, 12
- 8D may be served from hubs 4, 7, 12
- 8E may be served from hubs 4, 7, 12
- 8F may be served from hubs 4, 7, 12
- 8H1 may be served from hubs 4, 7, 12
- 8H2 may be served from hubs 4, 7, 12
- 10H1 takes drops at its ground pad until minute 70
- 10H2 takes drops at its ground pad until minute 70
- 11H1 takes drops at its ground pad until minute 86
- 11H2 takes drops at its ground pad until minute 86
- 12H1 takes drops at its ground pad until minute 86
- 12H2 takes drops at its ground pad until minute 86
- 15H1 takes drops at its ground pad until minute 147
- 16H1 takes drops at its ground pad until minute 149
- 16H2 takes drops at its ground pad until minute 86
- 1H1 takes drops at its ground pad until minute 120
- 1H2 takes drops at its ground pad until minute 120
- 2H1 takes drops at its ground pad until minute 70
- 2H2 takes drops at its ground pad until minute 70
- 3H1 takes drops at its ground pad until minute 120
- 3H2 takes drops at its ground pad until minute 120
- 5H1 takes drops at its ground pad until minute 70
- 5H2 takes drops at its ground pad until minute 70
- 6H1 takes drops at its ground pad until minute 70
- 6H2 takes drops at its ground pad until minute 70
- 7H1 takes drops at its ground pad until minute 70
- 7H2 takes drops at its ground pad until minute 70
- 8H1 takes drops at its ground pad until minute 116
- 8H2 takes drops at its ground pad until minute 116

Orders, each a JSON object with one key:
- {"reserve": {"hub": H, "count": N, "site": "CODE",
     "until_min": M}}: keep N vehicles at hub H for that site
     only.
- {"avoid": {"area": {"circle": {"center_km": [x, y],
     "radius_km": r}}, "until_min": M}} or {"avoid": {"area":
     {"corridor": {"from_km": [x, y], "to_km": [x, y],
     "half_width_km": w}}, "until_min": M}}: route every
     flight around the area.
- {"reposition": {"class": "fixed_wing" or "multirotor" or
     null, "count": N, "from_hub": A, "to_hub": B,
     "until_min": M or null}}: fly N idle vehicles from hub A
     to hub B, which becomes their hub; with until_min they
     fly home once idle after that minute.
- {"transfer": {"item": "O-", "units": N, "from_hub": A,
     "to_hub": B}}: fly N units of stock from hub A to hub B.
- {"ground": {"class": "fixed_wing" or "multirotor" or null,
     "hubs": [H, ...], "until_min": M}}: no launches of that
     class (null: any) from those hubs.
- {"prioritize": {"site": "CODE" or null, "urgency": "trauma"
     or "urgent" or "routine" or null, "until_min": M}}:
     serve those requests first.
- {"serve_from": {"site": "CODE", "hubs": [H, ...]}}: let
     those hubs serve the site besides its own.
- {"use_pad": {"site": "CODE", "until_min": M}}: deliver to
     that hospital's ground pad by drop instead of landing on
     its rooftop.

Reply with {"orders": [...]} and nothing else; {"orders": []}
     changes nothing.
```

The reply:

```
{"orders": [{"use_pad": {"site": "9H1", "until_min": 155}},
     {"reposition": {"class": "fixed_wing", "count": 2,
     "from_hub": 5, "to_hub": 3, "until_min": 120}},
     {"reposition": {"class": "fixed_wing", "count": 1,
     "from_hub": 7, "to_hub": 1, "until_min": 120}}]}
```

A veto prompt has the same head and, instead of the order format, the replanner's numbered proposals and a one-line instruction. The tail of one such prompt to Haiku, at minute 88 on a 16-hub day after a stock shortage, and its reply: the replanner proposed three stock transfers, each with lending permissions for the receiving hub's sites, and the model dropped the third transfer and its permissions while keeping the other two.

```
Proposed orders from the dispatcher's planner:
1. transfer 50 units of O+ from hub 11 to hub 16
2. serve 16A from hubs 11
3. serve 16B from hubs 11
4. serve 16C from hubs 11
5. serve 16D from hubs 11
6. serve 16E from hubs 11
7. serve 16F from hubs 11
8. serve 16H1 from hubs 11
```

```
9. serve 16H2 from hubs 11
10. transfer 50 units of O+ from hub 8 to hub 12
11. serve 12A from hubs 8
12. serve 12B from hubs 8
13. serve 12C from hubs 8
14. serve 12D from hubs 8
15. serve 12E from hubs 8
16. serve 12F from hubs 8
17. serve 12H1 from hubs 8
18. serve 12H2 from hubs 8
19. transfer 50 units of O+ from hub 11 to hub 15
20. serve 15A from hubs 11
21. serve 15B from hubs 11
22. serve 15C from hubs 11
23. serve 15D from hubs 11
24. serve 15E from hubs 11
25. serve 15F from hubs 11
26. serve 15H1 from hubs 11
27. serve 15H2 from hubs 11

Reply with {"drop": [numbers]}, the proposed orders to drop,
     and nothing else; {"drop": []} keeps them all.
```

```
{"drop": [19,20,21,22,23,24,25,26,27]}
```

IV. Per-seed scores

The paper's model table gives means. Table I gives the on-time score and the moves made (kept, for the veto) on each seed: seeds 1 to 3 at 16 hubs and 1 and 2 at 64. On the same seeds local rules score 0.69, 0.78, and 0.75 at 16 hubs and 0.73 and 0.70 at 64; the replanner 0.68, 0.74, and 0.73, and 0.62 and 0.62.

V. Luna through the API and command-line client

We reran Luna through the OpenAI API on the same three seeds at 16 hubs and two at 64, in both roles. The mission, study system prompt, exceptions view, and feedback setting were unchanged. The API supplies no tools; the earlier Codex command-line client retained coding-agent tools. Table II compares the means. The paper's model table and this appendix's per-seed scores and counts use the API runs.

The API does not consistently improve the score. At 64 hubs it improves the live manager's mean but lowers the veto's; both roles make more moves. All ten API runs score below local rules on their matched seeds. Five live-manager replies could not be parsed; all veto replies were readable. These few reruns compare the two client paths, not tool availability in isolation, and do not separate client effects from model sampling.

VI. Interface and policy checks

Table III gives the two 16-hub seeds behind the interface comparison in the paper. Each cell is the on-time score and number of applied moves. The checker and the donor-state line are separate changes to the outcome-feedback setting; the reserve row also uses the checker. On these seeds, local rules score 0.69 and 0.78, and the unguarded replanner scores 0.68 with 23 moves and 0.74 with 27.

With the same reserve guard and no model, the replanner scores 0.69 with 20 moves on seed 1 and 0.74 with 21 on seed 2.

Table IV gives the policies Haiku and Sonnet wrote before the day, and their single revisions after seeing two day summaries. The local-rules scores on the same three seeds are 0.69, 0.78, and 0.75.

Table V gives the quiet-day check, with every event kind at a quarter of its usual rate. The model rows use hub summaries and outcome feedback.

VII. Surge length

The paper summarizes the surge-length means. Table VI gives each seed: one scripted surge at hub 6 and its four neighbors on an otherwise quiet 16-hub day, at 16 trauma requests per hub per 15 minutes, held for 15, 60, 120, and 240 minutes.

VIII. Per-run counts

Table VII gives, for each of the 40 model runs in the paper's model table, the number of model calls, the orders the model gave (or, for the veto, the replanner's proposals it kept), how many were applied and how many refused by the dispatcher, the replies that could not be read, the moves (applied relocations and stock flights), the lending permissions, and the vehicles lost.

TABLE I

PER-SEED ON-TIME SCORE AND MOVES FOR THE 40 MODEL RUNS, IN SEED ORDER WITHIN EACH CELL.

Hubs	Role	Haiku	Sonnet	Opus	GPT-6 Luna
16	Live	0.64 / 0.74 / 0.73, 7 / 7 / 10	0.61 / 0.66 / 0.60, 36 / 48 / 56	0.72 / 0.69 / 0.66, 85 / 71 / 54	0.59 / 0.62 / 0.51, 2 / 6 / 1
16	Veto	0.68 / 0.76 / 0.69, 16 / 18 / 23	0.67 / 0.74 / 0.74, 19 / 34 / 23	0.67 / 0.73 / 0.74, 20 / 30 / 20	0.69 / 0.74 / 0.74, 18 / 25 / 23
64	Live	0.66 / 0.65, 61 / 67	0.60 / 0.57, 296 / 306	0.55 / 0.58, 623 / 339	0.63 / 0.61, 72 / 39
64	Veto	0.64 / 0.62, 134 / 169	0.63 / 0.62, 157 / 184	0.65 / 0.66, 158 / 133	0.62 / 0.63, 166 / 162

TABLE II

LUNA ON MATCHED SEEDS THROUGH THE COMMAND-LINE CLIENT (CLI) AND API. SCORES ARE WEIGHTED ON-TIME SHARES; MOVES ARE MEAN APPLIED RELOCATION AND STOCK-TRANSFER ORDERS. CLI SCORES RETAIN THE ORIGINAL TABLE'S ROUNDING.

		Score		Moves	
Hubs	Role	CLI	API	CLI	API
16	Live	0.62	0.57	3.3	3.0
16	Veto	0.72	0.72	13.7	22.0
64	Live	0.55	0.62	22.0	55.5
64	Veto	0.67	0.62	88.5	164.0

TABLE III
INTERFACE CHECK: ON-TIME SCORE / MOVES, SEEDS 1 AND 2.

View and checks	Haiku 1	Haiku 2	Sonnet 1	Sonnet 2
Hub summaries	0.66 / 52	0.70 / 44	0.66 / 66	0.68 / 37
Summaries + feedback	0.68 / 51	0.75 / 27	0.63 / 72	0.65 / 46
Feedback + checker	0.68 / 42	0.75 / 23	0.61 / 70	0.64 / 58
Feedback + donor state	0.62 / 58	0.73 / 32	0.62 / 74	0.69 / 47
Feedback + checker + reserve	0.70 / 25	0.76 / 10	0.65 / 38	0.68 / 28
Exceptions + feedback	0.64 / 7	0.74 / 7	0.61 / 36	0.67 / 48

TABLE IV
MODEL-WRITTEN POLICIES: ON-TIME SCORE / MOVES BY SEED.

Policy	Seed 1	Seed 2	Seed 3
Haiku	0.66 / 62	0.71 / 60	0.65 / 71
Haiku revised	0.66 / 62	0.71 / 60	0.65 / 71
Sonnet	0.63 / 67	0.69 / 60	0.65 / 71
Sonnet revised	0.63 / 36	0.69 / 32	0.65 / 39

TABLE V
QUIET 16-HUB DAYS: ON-TIME SCORE / MOVES BY SEED.

Manager	Seed 1	Seed 2
None	0.90 / 0	0.92 / 0
Local rules	0.92 / 0	0.94 / 0
Haiku	0.90 / 22	0.93 / 17
Sonnet	0.88 / 37	0.90 / 38

TABLE VI
PER-SEED ON-TIME SCORE BY SURGE LENGTH AND POLICY.

Surge (min)	Policy	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5
15	Local rules	0.947	0.928	0.937	0.938	0.939
	Lending	0.960	0.957	0.947	0.962	0.958
	Relocating	0.855	0.901	0.900	0.858	0.852
	Replanner	0.870	0.934	0.918	0.876	0.872
	Return	0.960	0.958	0.949	0.953	0.957
60	Local rules	0.657	0.651	0.649	0.645	0.656
	Lending	0.699	0.695	0.687	0.687	0.709
	Relocating	0.642	0.644	0.632	0.613	0.629
	Replanner	0.721	0.705	0.654	0.676	0.675
	Return	0.778	0.720	0.684	0.727	0.743
120	Local rules	0.403	0.382	0.385	0.393	0.384
	Lending	0.436	0.414	0.433	0.422	0.424
	Relocating	0.445	0.390	0.393	0.416	0.407
	Replanner	0.486	0.448	0.426	0.457	0.459
	Return	0.517	0.454	0.439	0.483	0.469
240	Local rules	0.265	0.256	0.253	0.263	0.256
	Lending	0.240	0.235	0.244	0.236	0.235
	Relocating	0.277	0.248	0.242	0.235	0.240
	Replanner	0.285	0.213	0.224	0.245	0.239
	Return	0.266	0.213	0.222	0.234	0.226

TABLE VII

PER-RUN COUNTS FOR THE 40 MODEL RUNS. REFUSED ORDERS ARE THOSE THE DISPATCHER COULD NOT APPLY, ALMOST ALL FOR TOO FEW IDLE VEHICLES. UNREADABLE REPLIES WERE NOT JSON IN THE EXPECTED FORM AND CHANGED NOTHING.

Hubs	Role	Model	Seed	Calls	Orders	Applied	Refused	Unreadable	Moves	Lends	Lost
16	Live	Haiku	1	49	94	90	4	0	7	45	5
16	Live	Haiku	2	50	69	65	4	0	7	18	3
16	Live	Haiku	3	53	120	117	3	1	10	70	0
16	Live	Sonnet	1	49	171	144	27	2	36	76	4
16	Live	Sonnet	2	50	137	129	8	1	48	47	9
16	Live	Sonnet	3	53	183	169	14	0	56	80	12
16	Live	Opus	1	49	385	326	59	0	85	192	5
16	Live	Opus	2	50	382	336	46	0	71	189	10
16	Live	Opus	3	53	409	379	30	0	54	250	4
16	Live	GPT-6 Luna	1	49	242	235	7	0	2	193	0
16	Live	GPT-6 Luna	2	50	235	226	9	1	6	164	1
16	Live	GPT-6 Luna	3	53	266	260	6	1	1	183	0
16	Veto	Haiku	1	47	205	205	0	0	16	152	0
16	Veto	Haiku	2	48	225	225	0	0	18	168	0
16	Veto	Haiku	3	46	232	232	0	0	23	172	0
16	Veto	Sonnet	1	46	244	244	0	0	19	184	0
16	Veto	Sonnet	2	46	156	155	1	0	34	79	0
16	Veto	Sonnet	3	46	152	151	1	0	23	88	0
16	Veto	Opus	1	46	213	213	0	0	20	152	0
16	Veto	Opus	2	46	145	144	1	0	30	72	0
16	Veto	Opus	3	48	190	189	1	0	20	128	0
16	Veto	GPT-6 Luna	1	47	289	289	0	0	18	232	0
16	Veto	GPT-6 Luna	2	46	237	235	2	0	25	168	0
16	Veto	GPT-6 Luna	3	46	315	314	1	0	23	250	0
64	Live	Haiku	1	185	362	335	27	0	61	121	34
64	Live	Haiku	2	185	348	321	27	2	67	98	21
64	Live	Sonnet	1	185	876	713	163	3	296	264	48
64	Live	Sonnet	2	185	989	793	196	6	306	329	56
64	Live	Opus	1	185	1843	1573	270	0	623	496	49
64	Live	Opus	2	185	1892	1651	241	0	339	899	47
64	Live	GPT-6 Luna	1	185	1577	1481	96	1	72	1096	11
64	Live	GPT-6 Luna	2	185	1450	1394	56	2	39	1000	4
64	Veto	Haiku	1	184	1060	1041	19	0	134	774	0
64	Veto	Haiku	2	184	1195	1165	30	0	169	848	0
64	Veto	Sonnet	1	182	888	871	17	0	157	552	0
64	Veto	Sonnet	2	184	924	891	33	0	184	544	0
64	Veto	Opus	1	182	1029	1019	10	0	158	696	0
64	Veto	Opus	2	184	1051	1035	16	0	133	736	0
64	Veto	GPT-6 Luna	1	182	1508	1471	37	0	166	1144	0
64	Veto	GPT-6 Luna	2	184	1480	1447	33	0	162	1120	0